Behavior and profile of an individual

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1. Introduction :

Reading the work of the French philosopher Ruwen Ogien about thought experiences (Ogien, 2011), I wondered if we could predict theoretical and real behavior from an individual's profile. «Thought experiences» are devices of the imagination. They are employed for various purposes such an entertainment, education, conceptual analysis, exploration, etc. (Brown, James Robert and Yiftach Fehige, 2023). Nevertheless, we focus here solely on the philosophical and psychological field of thought experiences. We therefore seek to determine whether there is a link between an individual's profile and their behavior in the face of fictitious and real situations. We also observe whether theoretical behavior reflects the actual behavior of an individual.

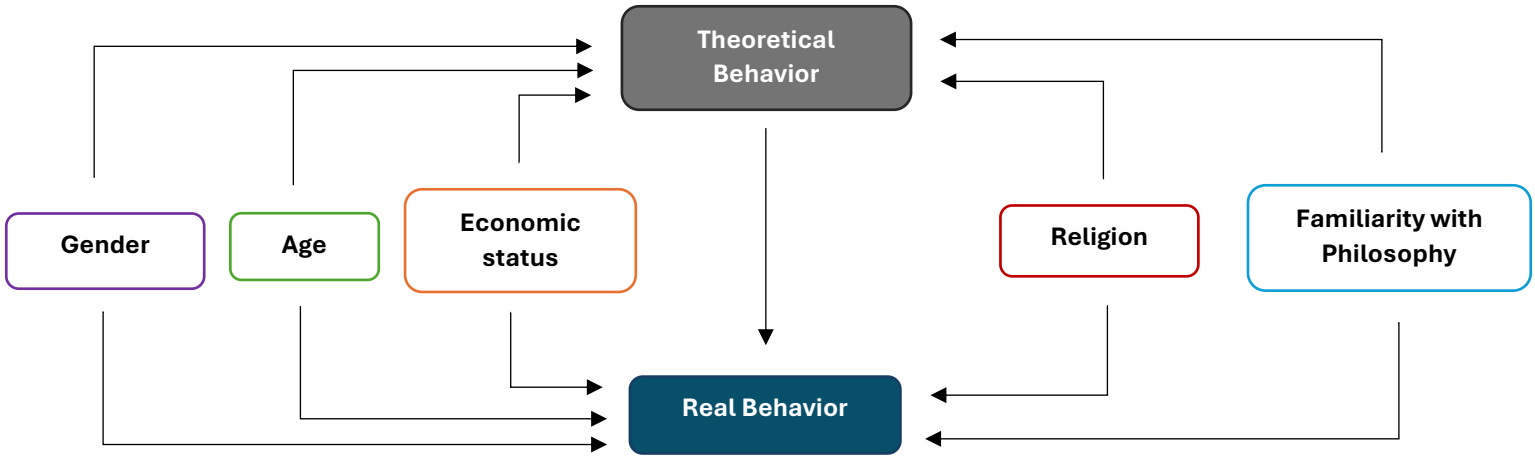


Figure 1. Directed acyclic graph representing the hypothesized causal links between age, gender, religious status, economic status, familiarity with philosophy, theoretical behavior and real behavior.
We anticipate that if this is the case, we should observe:

- **Prediction 1** : Theoretical behavior is influenced by an individual's profile
- **Prediction 2** : Real behavior is influenced by an individual's profile
- **Prediction 3** : Theoretical behavior reflects the real behavior

2. Data

The dataset comes from a questionnaire with 114 answers. It contains information on the respondent's profile (Age, Country of origin, Level of studies, Religion, Familiarity with philosophy, Gender, see **Figure 2**) as well as his responses to different thought experiences (Jarvis Thomson, 1971 and Robert Nozick, 1974). The individual then responds to concrete cases, such as his opinion on abortion and his disappointment at the discovery of a lie based on good intentions. For our problem, we choose not to study the level of education of the respondent and his country of origin. {Insert l'URL du questionnaire quand il sera publié pour montrer les énoncés des expériences des pensées}

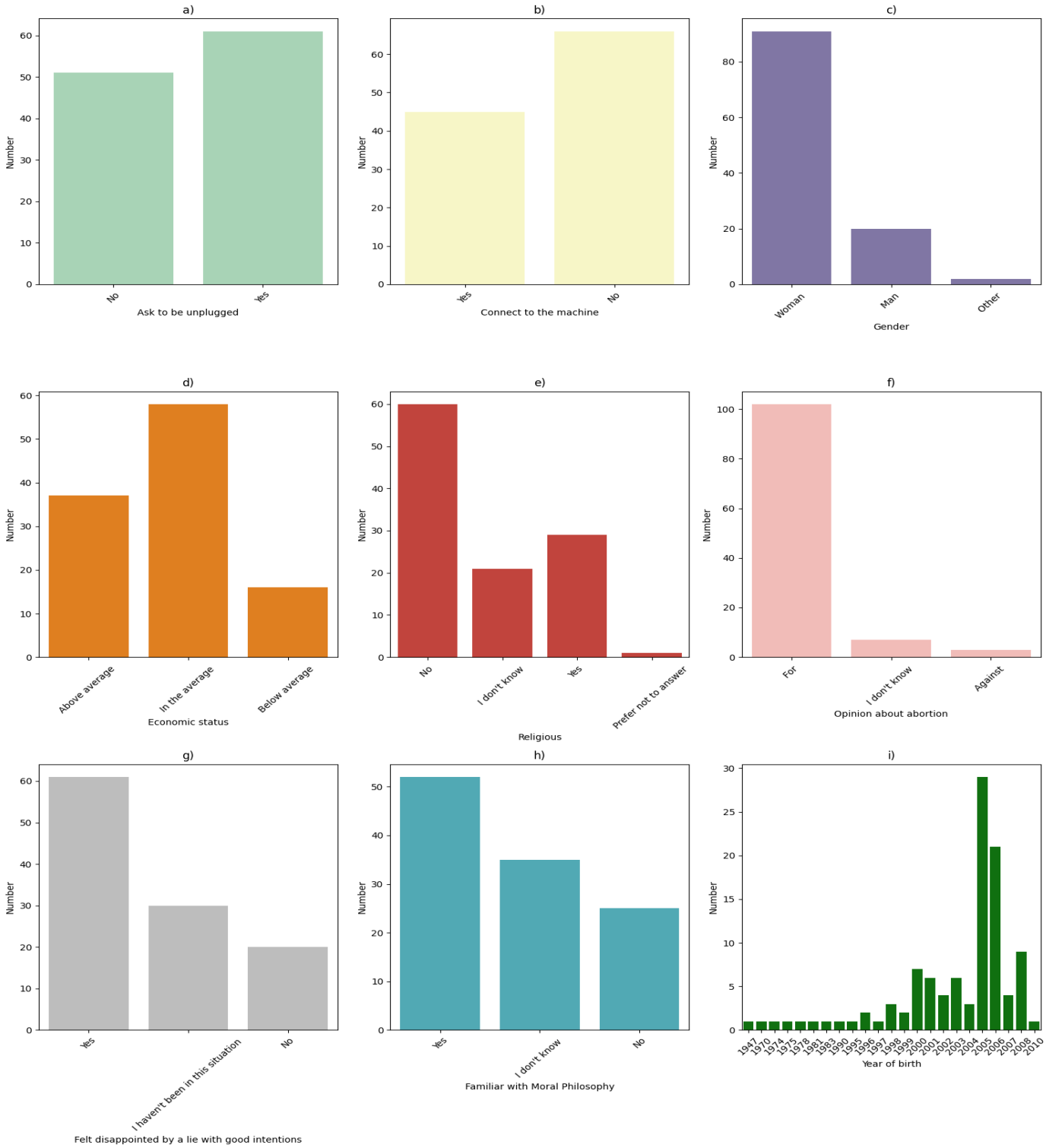


Figure 2.

a) Distribution of answers for the thought experience with the violinist, **b)** Distribution of answers for the thought experience with the machine, **c)** Distribution of respondents' gender , **d)** Distribution of respondents' economic status, **e)** Distribution of respondents' religious status, **f)** Distribution of respondents' opinion about abortion, **g)** Distribution of people feeling disappointed by a lie with good intentions, **h)** Distribution of respondents' familiarity with moral philosophy, **i)** Distribution of respondents' year of birth

3. Methods

To test prediction 1 and prediction 2, we performed a multivariate logistic regression to each of the predictions to analyze whether we can predict a respondent's answer if we know about their gender, religious status, economic status and familiarity with moral philosophy.

$$Th1_i \sim \beta_0 + \beta_1\ G_i + \beta_2\ R_i + \beta_3\ E_i + \beta_4 P_i + \beta_4 B_i + \mathcal{E}_i$$

$$Th2_i \sim \beta_0 + \beta_1\ G_i + \beta_2\ R_i + \beta_3\ E_i + \beta_4 P_i + \beta_4 B_i + \mathcal{E}_i$$

$$Rb1_i \sim \beta_0 + \beta_1\ G_i + \beta_2\ R_i + \beta_3\ E_i + \beta_4 P_i + \beta_4 B_i + \beta_5 Th1_i + \mathcal{E}_i$$

$$Rb2_i \sim \beta_0 + \beta_1\ G_i + \beta_2\ R_i + \beta_3\ E_i + \beta_4 P_i + \beta_4 B_i + \beta_5\ Th2_i + \mathcal{E}_i$$

With :

$Th1i$	Answers for the theoretical behavior 1 (Violinist) i
$Th2i$	Answers for the theoretical behavior 2 (Machine of happiness) i
Gi	Gender of the respondent i
Ri	Religious status of the respondent i
Ei	Economic status of the respondent i
Pi	Familiarity with moral philosophy of the respondent i
Bi	Year of birth of the respondent i
$\mathcal{E}i$	Error term draw in a normal distribution i

To test prediction 3, we performed χ^2 tests in relation with the following formula :

Opinion about Abortion ~ Answer for the Violinist's thought experience

Felt disappointed by a lie with good intentions ~ Answer for the machine of happiness's thought experience

4. Results

It would seem ($R^2 = 0.151$) that the economic status of an individual does allow us to predict his theoretical behavior in the face of the experience of thought of the violinist significantly (see **A1** and **A3**, *β =-1.044 p= 0.030)* , the same for gender (see **A1** and **A3**, *β =2.200 p= 0.009)*. However we can't predict the answer based only on the religious status (see **A1** and **A3**, *β =-0.0458 p= 0.143 > 0.05)*, age (see **A1** and **A3**, *β =-0.01 p=0.960 > 0.05)*, as well as his familiarity with moral philosophy (see **A1** and **A3**, *β =-1.078 p= 0.140)*.

It would seem (R^2 = 0.039) that the economic status of an individual does not allow us to predict his theoretical behavior in the face of the experience of thought of the machine of happiness significantly (see **A2** and **A4**, *β =-0.442, p= 0.290 > 0.05)*, the same for gender (see **A2** and **A4**, *β =-0.344, p=0.632 > 0.0)*, religious status (see **A2** and **A4**, *β =-0.250, p=0.387 > 0.05)*, age (see **A2** and **A4**, *β =-0.020, p=0.481 > 0.05)*, as well as his familiarity with moral philosophy (see **A2** and **A4**, *β =-0.400, p=0.550 > 0.05)*.

Moreover, it would seem that we cannot predict the response to the real case regarding opinion about abortion from the response to the violinist's thought experience (*p = 0.360 > 0.05)*, similarly to predict the response regarding disappointment in the face of a lie with good intentions from the response to the machine's thought experience to happiness (*p = 0.484 > 0.05)*.

5. Conclusion

We cannot predict the behavior of an individual in real and theoretical situations only through his profile, except for certain cases, such as gender or economic status for the thought experience of the violinist of Judith Thomson. We may plan to expand the scope of respondents to the questionnaire in order to check if this is still the case with a wider diversity in responses.

Moreover, it would seem that we cannot predict the response to the real case regarding opinion about abortion from the response to the violinist's thought experience of Judith Thomson, similarly to predict the response regarding disappointment in the face of a lie with good intentions from the response to the machine of happiness thought experience of Robert Nozick. This may mean that the thought experiences are not similar enough to the actual cases proposed, which would explain this difference in respondents' answers.

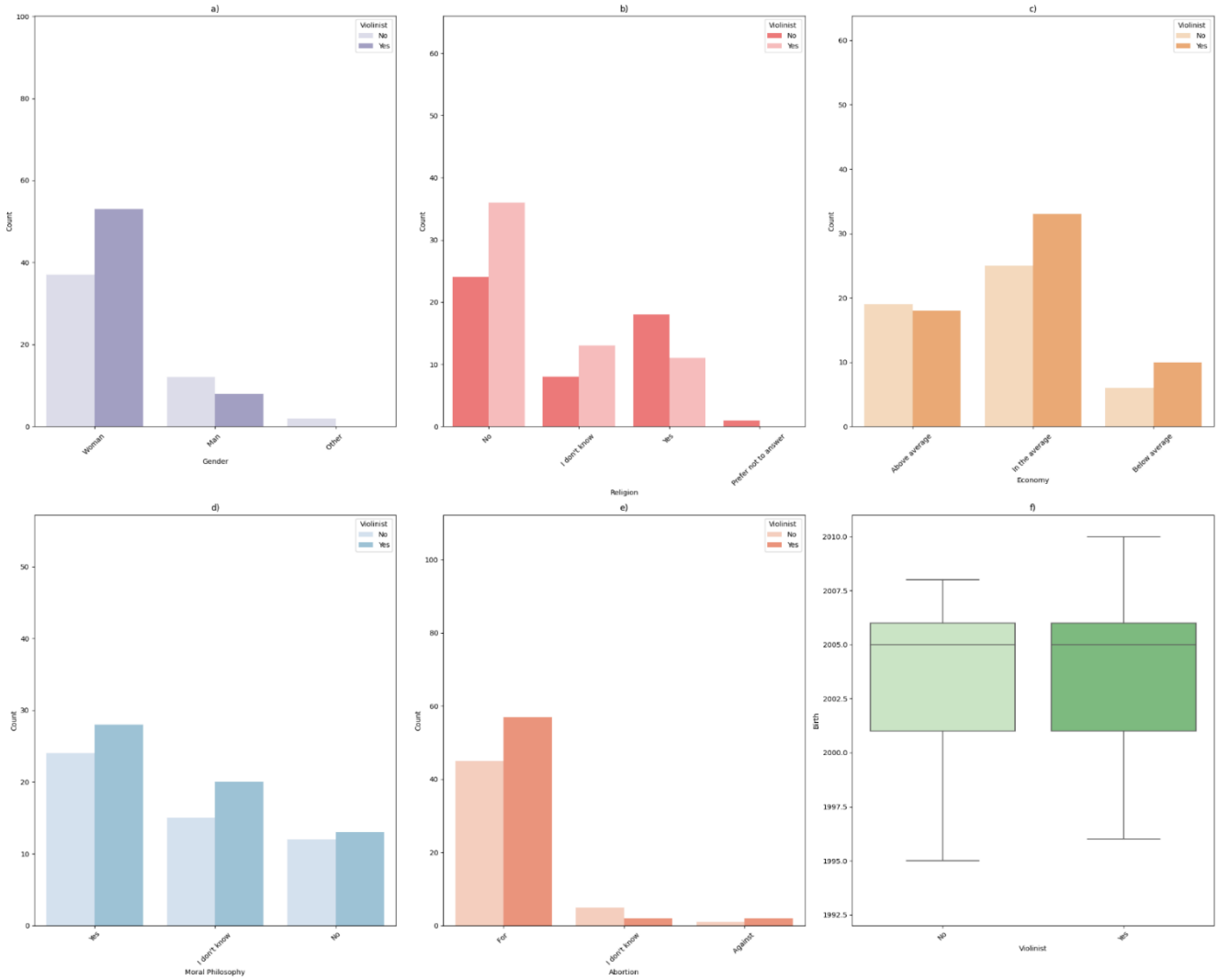
References :

Brown, James Robert and Yiftach Fehige, "Thought Experiments", *The Stanford Encyclopedia of Philosophy* (Winter 2023 Edition), Edward N. Zalta & Uri Nodelman (eds.), URL = <https://plato.stanford.edu/archives/win2023/entries/thought-experiment/>.

Thomson, Judith Jarvis (1971). "*A Defense of Abortion*". *Philosophy and Public Affairs.* 1 (1): 47–66. ISSN 1088-4963. JSTOR 2265091.

Nozick, Robert (2013). *Anarchy, State, and Utopia*. Basic Books. *ISBN* 978-0-465-05100-7.

Annex



A1.

a) Catplot of responses to the violinist's thought experience according to the gender of the respondents

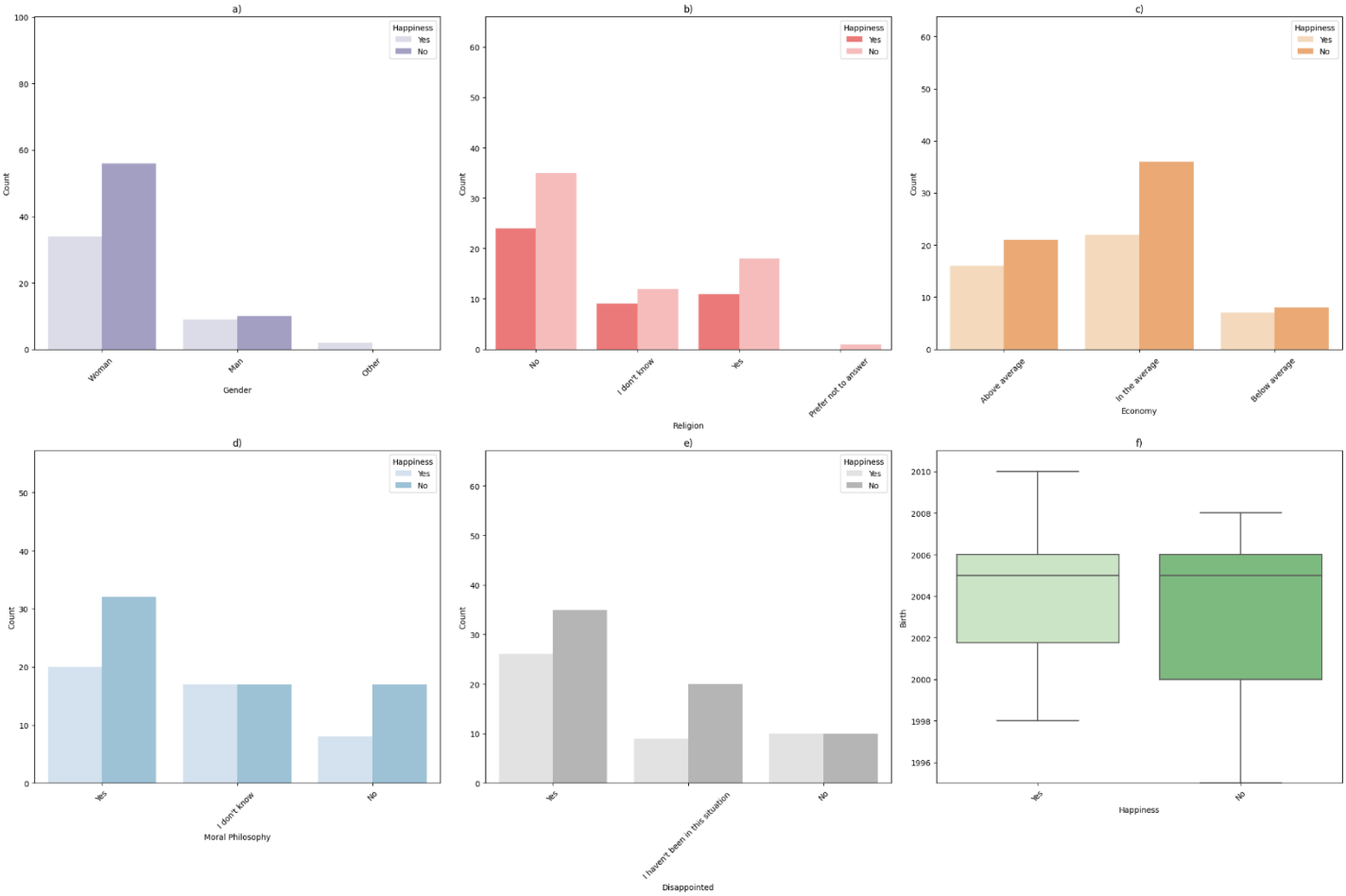
b) Catplot of responses to the violinist's thought experience according to the religious status of the respondents

c) Catplot of responses to the violinist's thought experience according to the economic status of the respondents

d) Catplot of responses to the violinist's thought experience according to the familiarity with moral philosophy of the respondents

e) Catplot of responses to the violinist's thought experience according to the opinion about abortion of the respondents

f) Boxplot of responses to the violinist's thought experience according to the year of birth of the respondents



A2.

a) Catplot of responses to the machine's thought experience according to the gender of the respondents

b) Catplot of responses to the machine's thought experience according to the religious status of the respondents

c) Catplot of responses to the machine's thought experience according to the economic status of the respondents

d) Catplot of responses to the machine's thought experience according to the familiarity with moral philosophy of the respondents

e) Catplot of responses to the machine's thought experience according to the response to the violinist's thought experience of the respondents

f) Boxplot of responses to the machine's thought experience according to the year of birth of the respondents

	coef	p-value
Economy	-1.044	0.030
Gender	2.200	0.009
Religion	-0.458	0.143
Familiarity with moral philosophy	-1.078	0.140
Age	-0.01	0.960

A3. Table of multivariate logistic regression of responses to the violinist's thought experience according to its variables confusing with in red the insignificant results and in green the significant results

	coef	p-value
Economy	-0.442	0.290
Gender	-0.344	0.632
Religion	-0.250	0.387
Familiarity with moral philosophy	0.400	0.550
Age	-0.020	0.481

A4. Table of multivariate logistic regression of responses to the machine's experience of thinking about happiness according to its variables confusing with in red the insignificant results and in green the significant results